# Additional case for GPK(Graham, Knuth and Patashnik) sequences

Aubert Voalaza Mahavily Romuald

**Objective:** To give the generating functions and study the asymptotic behavior of $T(n,k)$ defined by

$$T(n,k) = (a_2 n + a_1 k + a_0)\, T(n-1,k) + (b_2 n + b_1 k + b_0)\, T(n-1,k-1),$$

with the usual initial condition

$$T(0,0) = 1, \quad T(n,k) = 0 \text{ if } n < \max(k,0). \tag{1}$$

These sequences generalize many combinatorial numbers, such as Stirling numbers, Eulerian numbers, Lah numbers, and Whitney numbers. Graham, Knuth, and Patashnik posed this problem in their book [4]. Many years ago, Wilf [7] (and also Neuwirth [5]) found the generating function for the case $b_2 = 0$, which yields an explicit formula. Other cases remain open, both in terms of explicit formulas and asymptotics. For families generalizing Eulerian numbers, Hwang et al. [3] made a significant step using probabilistic methods to derive asymptotics in some cases as well. Hitczenko [8] did recently (2025) some cases using the generating function of $b_2 = 0$.

Here, we treat the case $a_2 = -a_1$ using context-grammar (Willian Chen sens) with Dumont tree-like structure associated (also system of differential equation combinatorial). We get the generating function and we try to give the asymptotic estimates and distribution of the random variables associated to $(T(n,k))$, using the basic theory in Flajolet [2].

**I would like to emphasize the following** This is not a piece of work intended for publication at this stage. It is simply part of my personal exercises on asymptotic methods that I studied earlier. Therefore, it has not undergone any formal verification, and I am not sure to what extent it contains new results or overlaps with existing work. It could potentially be combined with our paper, in preparation on context-free grammars (available on arXiv). This is a preliminary draft.

## 1 Generating functions for $a_1 + a_2 = 0$ (additional case)

We get these generating functions. The idea is to consider the grammar

$$\{x \to G(x) = (b_1+b_2)x^2 + (a_1+a_2)xy; \quad y \to G(y) = b_2 xy + a_2 y^2; \quad z \to G(z) = (b_1+b_2+b_0)zx + (a_0+a_2)zy\} \tag{2}$$

and use the following provable fact. If $\mathcal{D} = G(x)\partial_x + G(y)\partial_y + G(z)\partial_z$ then

$$\mathcal{D}^n(z) = z \sum_{k=0}^{n} T(n,k) x^k y^{n-k}. \tag{3}$$

We apply this fact in the Dumont [1] (Randrianirina [9]) point of views. What we use here, can be summarized as the following.

**Proposition 1.1.** *Let $\vec{y}(t) = (y_i(t))_{i=1,\ldots,v}$ be the solution of the analytic differential system*

$$y_i'(t) = G_i(\vec{y}(t)), \qquad y_i(0) = x_i, \quad i = 1, \ldots, v. \tag{4}$$

*Let $\mathcal{D} = \sum_{i=1}^{v} G(\vec{x}(t))\partial_{x_i}$. Then for each $i$, $\mathrm{Gen}(x_i,t) := \sum_{n\ge 0} \mathcal{D}^n(x_i)\frac{t^n}{n!} = y_i(t)$.*

In another hand, the data of the grammar $G$, the differential operator $\mathcal{D}$ and the differential system are equivalent. Here, we use only the analytic part of the differential system.
So firstly, we solve then the system of differential equation associated to the grammar (2).

$$\begin{cases} X'(t) &= (b_1+b_2)X^2, \quad X(0)=x, \\ Y'(t) &= b_2XY + a_2Y^2, \quad Y(0)=y, \\ Z'(t) &= (b_0+b_2)ZX + (a_0+a_2)ZY, \quad Z(0)=z \end{cases} \tag{5}$$

Applying proposition 1.1 with the fact (3), one can get

$$\sum_{n\geq 0} z \sum_{k=0}^{n} T(n,k)x^k y^{n-k}\frac{t^n}{n!} = \sum_{n\geq 0} \mathcal{D}^n(z)\frac{t^n}{n!} = Z(t,x,y,z).$$

We try to solve the system (5) and substitute $y=1$. So

$$F(t,x) = \frac{Z(t)}{z} = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!}. \tag{6}$$

....WILL WRITE THE CONCRETE (SHORT) WAY TO GET THE SOLUTION OF (5)....

In all cases, the generating functions can be derived explicitly because the system (5) can be solved using elementary functions. The explicit formulas are then obtained from these generating functions.

## 1.1 Case 1: $b_1+b_2=0$

### Case 1A: $a_2\neq 0$, $b_2=0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt)\,(1-a_2t)^{-\frac{a_0+a_2}{a_2}}.$$

Developing the generating function, we obtain

$$T(n,k) = \binom{n}{k} b_0^k\, a_2^{n-k}\, \left(\frac{a_0+a_2}{a_2}\right)_{n-k}.$$

### Case 1B: $a_2=0$, $b_2=0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt + a_0t)\,.$$

This is trivial, even without passing through generating functions. $T(n,k)=\binom{n}{k}a_0^k b_0^{n-k}$.

### Case 1C: $a_2\neq 0$, $b_2\neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\big(b_0x\,t\big)\left(\frac{b_2x}{b_2x+a_2-a_2\,e^{b_2xt}}\right)^{\frac{a_0+a_2}{a_2}}.$$

Expanding the generating function, we obtain

$$T(n,k) = \binom{-\frac{a_0+a_2}{a_2}}{n-k}\left(\frac{a_2}{b_2}\right)^{n-k}\sum_{j=0}^{n-k}(-1)^j\binom{n-k}{j}(b_0+jb_2)^n.$$

### Case 1D: $a_2 = 0$, $b_2 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} T(n,k)x^k \frac{t^n}{n!} = \exp\left( b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)\right).$$

Developing the generating function, we obtain

$$T(n,k) = \frac{(a_0/b_2)^{n-k}}{(n-k)!} \sum_{j=0}^{n-k} (-1)^{n-k-j} \binom{n-k}{j} (b_0 + j b_2)^n.$$

We can have also

$$T(n,k) = \left(\frac{a_0}{b_2}\right)^{n-k} \sum_{r=n-k}^{n} \binom{n}{r} b_0^{n-r} b_2^r S(r, n-k),$$

$S(r,m)$ is Stirling numbers of the second kind.

## 1.2 Case 2: $b_1 + b_2 \neq 0$, $a_2 = 0$

### Case 2A: $b_1 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$\begin{aligned} F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} T(n,k)x^k \frac{t^n}{n!} &= (1 - (b_1 + b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}} \\ &\times \exp\left[ -\frac{a_0}{b_1 x}\left( (1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}} - 1 \right)\right]. \end{aligned}$$

Expanding the generating function, we obtain

$$T(n,k) = \frac{n!}{(n-k)!}\left(-\frac{a_0}{b_1}\right)^{n-k} (-(b_1+b_2))^n \sum_{j=0}^{n-k} \binom{n-k}{j} (-1)^{(n-k)-j} \binom{\frac{b_1 j - (b_0+b_1+b_2)}{b_1+b_2}}{n}.$$

### Case 2B: $b_1 = 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = (1 - b_2 x t)^{-\frac{(b_0+b_2)x + a_0}{b_2 x}}.$$

Developing the generating function, we obtain

$$T(n,k) = a_0^{n-k} \sum_{1\leq m_1 < \cdots < m_k \leq n} \prod_{r=1}^{k} (b_0 + m_r b_2),$$

or

$$T(n,k) = (-1)^{n-k}\left(\frac{a_0}{b_2}\right)^{n-k} b_2^n \sum_{m=n-k}^{n} \binom{n}{m} s(m,k)\,(-1)^m \left(\frac{b_0+b_2}{b_2}\right)_{n-m}.$$

### 1.3 Case 3: $b_1+b_2\neq 0$, $a_2\neq 0$

#### Case 3A: $b_1\neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x)=\sum_{n\geq 0}\sum_{k=0}^{n}T(n,k)x^k\frac{t^n}{n!}=(1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$

$$\times\left(\frac{1}{1-\frac{a_2}{b_1x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]}\right)^{\frac{a_0+a_2}{a_2}}.$$

Developing the generating function, we obtain

$$T(n,k)=\binom{\alpha+n-k-1}{n-k}\left(\frac{a_2}{b_1}\right)^{n-k}n!\,(b_1+b_2)^n\sum_{j=0}^{n-k}(-1)^j\binom{n-k}{j}\binom{\frac{b_0+b_1+b_2-jb_1}{b_1+b_2}+n-1}{n}.$$

#### Case 3B: $b_1=0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x)=\sum_{n\geq 0}\sum_{k=0}^{n}T(n,k)x^k\frac{t^n}{n!}=(1-b_2xt)^{-\frac{b_0+b_1}{b_2}-1}$$

$$\times\left(\frac{1}{1+\frac{a_2}{b_2x}\ln(1-b_2xt)}\right)^{\frac{a_0+a_2}{a_2}}.$$

Then we obtain

$$\sum_{n\geq 0}T(n,k)\frac{t^n}{n!}=(1-b_2t)^{-\frac{b_0+b_1}{b_2}-1}\cdot\frac{(-1)^k\left(\frac{a_0+a_2}{a_2}\right)_k}{k!}\left(\frac{a_2}{b_2}\ln(1-b_2t)\right)^k.$$

## 2 Analytic approach

For this case we assume that $(T(n,k))\subset\mathbb{R}_{\geq 0}$.

As usual, we set the random variable $X_n$ defined by

$$\mathbb{P}(X_n=k):=\frac{T(n,k)}{\sum_{j=0}^{n}T(n,j)}.$$

Considering the polynom

$$P_n(x):=\sum_{k=0}^{n}T(n,k)x^k=n![t^n]F(t,x),$$

we have then $F(t,x)=\sum_{n\geq 0}P_n(x)\frac{t^n}{n!}$ and $X_n$ is the random variable of probability generating function

$$p_n(x)=\mathbb{E}(x^{X_n})=\frac{P_n(x)}{P_n(1)}=\frac{[t^n]F(t,x)}{[t^n]F(t,1)}.$$

We use also tools from complex analysis to estimate an polymorphic function.

**Definition 2.1** (Hayman–admissibility [2]). *Let $G(z)$ have radius of convergence $\rho$ with $0 < \rho \leq +\infty$ and be always positive on some subinterval $(R_0, \rho)$ of $(0, \rho)$. The function $G(z)$ is said to be admissible if it satisfies the following three conditions.*

- ***H***$_1$. *[Capture condition]* $\lim_{r \to \rho} b(r) = +\infty$.
- ***H***$_2$. *[Locality condition] For some function $\theta_0(r)$ defined over $(R_0, \rho)$ and satisfying $0 < \theta_0 < \pi$, one has*
$$G(re^{i\theta}) \sim G(r)e^{i\theta a(r) - \theta^2 b(r)/2} \quad \text{as } r \to \rho,$$
*uniformly in $|\theta| \leq \theta_0(r)$.*
- ***H***$_3$. *[Decay condition] Uniformly in $\theta_0(r) \leq |\theta| < \pi$,*
$$G(re^{i\theta}) = o\left(\frac{G(r)}{\sqrt{b(r)}}\right).$$

*Here*

$$a(r) = r\frac{G'(r)}{G(r)} = r(\log G(r))', \quad b(r) = r\frac{G'(r)}{G(r)} + r^2\frac{G''(r)}{G(r)} - r^2\left(\frac{G'(r)}{G(r)}\right)^2 = r(\log G(r))' + r^2(\log G(r))''.$$

**Theorem 2.1** (Coefficients of admissible functions [2]). *Let $G(z)$ be an $H$–admissible function and $\zeta \equiv \zeta(n)$ be the unique solution in the interval $(R_0, \rho)$ of the saddle point equation*

$$\zeta\frac{G'(\zeta)}{G(\zeta)} = n. \tag{48}$$

*The Taylor coefficients of $G(z)$ satisfy, as $n \to \infty$*

$$g_n \equiv [z^n]G(z) \sim \frac{G(\zeta)}{\zeta^n\sqrt{2\pi b(\zeta)}} \quad \text{as } n \to \infty \tag{49}$$

*with $b(z) = z^2 h''(z) + z h'(z)$ and $h(z) = \log G(z)$.*

**Theorem 2.2** (Generalized quasi-powers [2]). *Assume that, for $u$ in a fixed neighborhood $\Omega$ of 1, the generating function $p_n(u)$ of a non-negative discrete random variable $X_n$ admits a representation of the form*

$$p_n(u) = \exp\big(h_n(u)\big)\,(1 + o(1)),$$

*that holds uniformly, where each $h_n(u)$ is analytic in $\Omega$. Assume also the condition,*

$$h_n'(1) + h_n''(1) \longleftarrow +\infty \qquad \text{and} \qquad \frac{h_n'''(u)}{(h_n'(1) + h_n''(1))^{3/2}} \longrightarrow 0, \tag{55}$$

*uniformly for $u \in \Omega$. Then, the random variable*

$$X_n^* = \frac{X_n - h_n'(1)}{(h_n'(1) + h_n''(1))^{1/2}}$$

*converges in distribution to a normal law with parameters $(0, 1)$.*

**Theorem 2.3** (Algebraic singularity schema [2])**.** *Let $F(z,u)$ be analytic at $(0,0)$ with nonnegative coefficients. Assume the following conditions*

*(i)* ***Algebraic perturbation:*** *there exist three functions $A, B, C$, analytic in a domain.........*

$$F(z,u) = A(z,u) + B(z,u)C(z,u)^{-\alpha}. \tag{7}$$

*Assume $\rho$ is the unique simple root of $C(z,1) = 0$ in $|z| \leq r$, and $B(\rho,1) \neq 0$.*

*(ii)* ***Nondegeneracy.***

$$\partial_z C(\rho,1) \cdot \partial_u C(\rho,1) \neq 0.$$

*(iii)* ***Variability.***

$$\mathfrak{v}\Big(\frac{\rho}{\rho(x)}\Big) = -\frac{\rho''(1)}{\rho(1)} - \frac{\rho'(1)}{\rho(1)} + \left(\frac{\rho'(1)}{\rho(1)}\right)^2 \neq 0.$$

*Then*

$$p_n(u) = \frac{[z^n]F(z,u)}{[z^n]F(z,1)}$$

*converges to a Gaussian law with speed $O(n^{-1/2})$.*

## 2.1 Case 1A $a_1 = -a_2 \neq 0$, $b_1 = b_2 = 0$

From

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = \exp(b_0 x t)\,(1-a_2 t)^{-\frac{a_0+a_2}{a_2}},$$

we get

$$\sum_{n\geq 0} T(n,k)\frac{t^n}{n!} = \frac{(b_0 t)^k}{k!}(1-a_2 t)^{-\frac{a_0+a_2}{a_2}}.$$

Then

$$T(n,k) = \frac{n! b_0^k}{k!}[t^n]t^k(1-a_2 t)^{-\frac{a_0+a_2}{a_2}} = \frac{n! b_0^k}{k!}[t^{n-k}](1-a_2 t)^{-\frac{a_0+a_2}{a_2}}.$$

So if $k$ is not near to $n$ ($n-k$ is huge) and $n \to \infty$

$$T(n,k) \sim \frac{b_0^k}{k!\,\Gamma\left(\frac{a_0+a_2}{a_2}\right)}\, a_2^{n-k}\,(n-k)^{\frac{a_0}{a_2}}\,\sqrt{2\pi n}\left(\frac{n}{e}\right)^n. \tag{8}$$

We note that the asymptotic at (8) is sure only for $\frac{a_0+a_2}{a_2} \notin \mathbb{Z}_{\leq 0}$. Otherwise, $a_2$ divide $a_0$, opposite sign, let say $a_0 = -qa_0$, for $q \in \mathbb{N}$. So $(1-a_2 t)^{-\frac{a_0+a_2}{a_2}} = (1-a_2 t)^{q-1}$ is of degree $q-1$. Then $T(n,k) = 0$ unless $k \geq n+1-q$, where $T(n,k) = \frac{n! b_0^k}{k!}(-a_2^{n-k})\binom{\frac{a_0+a_2}{a_2}}{n-k}$.

**Proposition 2.1.** *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = (a_2 n - a_2 k + a_0)\,T(n-1,k) + b_0\,T(n-1,k-1).$$

*Assume $\frac{a_0+a_2}{a_2} \notin \mathbb{Z}_{\leq 0}$ and all coefficients are non-negative. $F(t,x)$ be the exponential bivariate generating function. Then The random variable $X_n$ of probability generating function*

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

*converges in distribution to a Poisson random variable.*

*Proof.* We write the asymptotic of $p_n(x)$. First, we know

$$[t^n](1-a_2t)^{-\alpha} = \frac{a_2^n}{\Gamma(\alpha)}\, n^{\alpha-1}\left(1+O\Big(\frac{1}{n}\Big)\right).$$

Also $e^{b_0xt}$ is analytic at the singularity $\rho=\frac{1}{a_2}$, then

$$[t^n]F(t,x) = e^{b_0x/a_2}\frac{a_2^n}{\Gamma(\frac{a_0+a_2}{a_2})}\, n^{\frac{a_0}{a_2}}\left(1+O\Big(\frac{1}{n}\Big)\right)$$

and

$$p_n(x) = \frac{e^{b_0x/a_2}}{e^{b_0/a_2}}\cdot\frac{1+O(1/n)}{1+O(1/n)}.$$

So we have

$$p_n(x) \longrightarrow \exp(\lambda(x-1))\,, \qquad \lambda = \frac{b_0}{a_2}.$$

□

## 2.2 Case 1B $a_1 = a_2 = 0$, $b_1 = b_2 = 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt + a_0t)\,.$$

This case is trivial, in many different ways, we can get quickly that

$$T(n,k) = \binom{n}{k}a_0^{n-k}b_0^k \qquad \Longrightarrow\; X_n \sim \mathrm{Bin}(n, \frac{b_0}{(a_0+b_0)}).$$

## 2.3 Case 1C: $a_1 = -a_2 \neq 0$, $b1 = -b_2 \neq 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\big(b_0x\,t\big)\left(\frac{b_2x}{b_2x + a_2 - a_2\,e^{b_2xt}}\right)^{\frac{a_0+a_2}{a_2}}.$$

## 2.4 Case 1D: $a_2 = 0$, $b_1 = -b_2 \neq 0$

$$T(n,k) = a_0\,T(n-1,k) + (b_2n - b_2k + b_0)\,T(n-1,k-1).$$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\right). \tag{9}$$

Equation (9) is equal to

$$\sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)(\frac{1}{x})^{n-k}\frac{(xt)^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\right)$$

$$\Longrightarrow \sum_{n\geq 0}\sum_{k=0}^{n} T(n,n-k)u^k\frac{z^n}{n!} = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)u^{n-k}\frac{z^n}{n!} = e^{b_0z}\exp\left(\frac{a_0}{b_2}\big(e^{b_2z}-1\big)u\right)$$

$$\Longrightarrow \sum_{n\geq 0} T(n,n-k)\frac{z^n}{n!} = \frac{a_0^k}{k!b_2^k}e^{b_0z}\big(e^{b_2z}-1\big)^k. \tag{10}$$

So

$$T(n,k) = (\frac{n!a_0^k}{k!b_2^k})[z^n]G_k(z), \qquad G_k(z) = e^{b_0z}\big(e^{b_2z}-1\big)^k. \tag{11}$$

Let's check quickly the admissibility of $G_k(z)$. It is always positive on $(R_0 = 0, \rho = +\infty)$. We compute first $a(r)$ and $b(r)$ and get

$$a(r) = r\left(b_0 + k\frac{b_2 e^{b_2 r}}{e^{b_2 r} - 1}\right), \quad b(r) = r\left(b_0 + k\frac{b_2 e^{b_2 r}}{e^{b_2 r} - 1}\right) - r^2 k\frac{b_2^2 e^{b_2 r}}{(e^{b_2 r} - 1)^2}.$$

Then for $b_0, b_2 > 0$ H1, capture condition is verified: $\lim_{n\to+\infty} b(r) = +\infty$. We expend $e^{b_0 r e^{i\theta}}$ and $(e^{b_1 r e^{i\theta}} - 1)$ wisely around $\theta = 0$ (small enough $\theta$) to get H2, locality condition

$$G(re^{i\theta}) \sim G(r)e^{i\theta a(r) - \theta^2 b(r)/2} \quad \text{as } r \to +\infty.$$

Let $c = (b_0 + kb_2)(1 - \cos(\theta_0))$ so that for $\theta_0(r) \le \theta \le \pi$, $(b_0 + kb_2)(\cos\theta - 1) \le -c$ and

$$\left|\frac{G(re^{i\theta})}{G(r)/\sqrt{b(r)}}\right| = \sqrt{b(r)}\, e^{(b_0+kb_2)r(\cos\theta-1)}\,(1 + o(1)) \le \sqrt{b(r)}\, e^{-cr} \longrightarrow 0, \quad (r \to \infty,\ |\theta| \ge \theta_0(r)).$$

So H3, decay condition is satisfied, thus $G_k$ is H-admissible. The saddle point equation is given by

$$\zeta_k(n)\left(b_0 + k\frac{b_2 e^{b_2\zeta_k(n)}}{e^{b_2\zeta_k(n)} - 1}\right) = n. \tag{12}$$

We will apply the Coefficients admissible theorem. Let's estimate first $\zeta_k(n)$

$$\zeta_k(n)\left(\frac{b_0}{n} + \frac{k}{n}\frac{b_2 e^{b_2\zeta(n)}}{e^{b_2\zeta(n)} - 1}\right) = 1.$$

So if $n \to \infty$ (we assume that $b_0$ is not very high comparing to $b_2$), we have

$$\zeta_k(n)\frac{k}{n}\frac{b_2 e^{b_2\zeta_k(n)}}{e^{b_2\zeta_k(n)} - 1} \sim 1 \quad \text{or} \quad \frac{b_2\zeta_k(n)}{1 - e^{-b_2\zeta_k(n)}} \sim \frac{n}{k}.$$

Therefore $\zeta_0(n) = n/b_0$ and for $k \ne 0$

$$\zeta_k(n) \sim \frac{1}{b_2}\left(\frac{n}{k} + W\left(-\frac{n}{k}e^{-\frac{n}{k}}\right)\right). \tag{13}$$

Thus the Coefficients admissible theorem gives us the following results.

**Proposition 2.2.** *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = a_0\,T(n-1,k) + (b_2 n - b2k + b_0)\,T(n-1,k-1).$$

*We have*

$$T(n,n-k) \sim \left(\frac{n!a_0^k}{k!b_2^k}\right)\frac{G_k(\zeta_k(n))}{\zeta_k(n)^n\sqrt{2\pi n\left(1 - \frac{b_2\zeta_k(n)}{(e^{b2\zeta_k(n)}-1)}\right)}}.$$

*Equivalently*

$$T(n,k) \sim \left(\frac{n!a_0^{n-k}}{(n-k)!b_2^{n-k}}\right)\frac{e^{b_0\zeta_{n-k}(n)}\left(e^{b_2\zeta_{n-k}(n)} - 1\right)^{n-k}}{\zeta_{n-k}(n)^n\sqrt{2\pi n\left(1 - \frac{b_2\zeta_{n-k}(n)}{(e^{b2\zeta_{n-k}(n)}-1)}\right)}},$$

*for $\zeta_{n-k}(n)$ given by* (12) *that can be as in* (13) *or $\zeta_{n-k}(n) \sim \frac{n}{b_0+(n-k)b_2}$, accordingly ($k \ll n$ or $n - k \ll n$) or not.*

Here $k \ll n$ is also depending on how big $b_0 \gg b_2$. Somehow, $k \ll n$ should be $kf(b_0) \ll ng(b_2)$ for some increasing functions $f$ and $g$.

| $(n, k)$ | Exact value | Approximate value | Relative error |
|---:|---:|---:|---:|
| $(20000, 5)$ | $1.245852907 \times 10^{18100}$ | $1.280350540 \times 10^{18100}$ | 0.0276906 |
| $(20000, 10)$ | $9.735206503 \times 10^{18134}$ | $9.732543012 \times 10^{18134}$ | 0.0002736 |
| $(20000, 50)$ | $1.001227349 \times 10^{18392}$ | $1.003427242 \times 10^{18392}$ | 0.0021962 |
| $(20000, 100)$ | $1.127882259 \times 10^{18692}$ | $1.129909884 \times 10^{18692}$ | 0.0017976 |
| $(20000, 300)$ | $1.697672189 \times 10^{19808}$ | $1.698279109 \times 10^{19808}$ | 0.0003574 |
| $(20000, 700)$ | $4.043920535 \times 10^{21871}$ | $4.044741395 \times 10^{21871}$ | 0.0002030 |
| $(20000, 1000)$ | $5.606744636 \times 10^{23340}$ | $5.607414282 \times 10^{23340}$ | 0.0001194 |
| $(20000, 1200)$ | $2.386834815 \times 10^{24294}$ | $2.387338510 \times 10^{24294}$ | 0.0002110 |
| $(20000, 1500)$ | $3.426108298 \times 10^{25693}$ | $3.426491193 \times 10^{25693}$ | 0.0001117 |
| $(20000, 1900)$ | $1.334384744 \times 10^{27510}$ | $1.334538680 \times 10^{27510}$ | 0.0001153 |
| $(20000, 1950)$ | $5.227751444 \times 10^{27733}$ | $5.228480286 \times 10^{27733}$ | 0.0001394 |
| $(20000, 1990)$ | $1.182081908 \times 10^{27912}$ | $1.182215891 \times 10^{27912}$ | 0.0001134 |
| $(20000, 1995)$ | $2.161821454 \times 10^{27934}$ | $2.162018694 \times 10^{27934}$ | 0.0000913 |

Table 1: Comparison of exact and approximate values with relative errors for $a_0 = 8, b_0 = 4, b_2 = 3$.

On another hand, from the Cauchy Integral formula, the equation (10) implies

$$T(n, k) = \frac{n!}{2\pi i} \frac{a_0^{n-k}}{(n-k)! b_2^{n-k}} \int_{\mathcal{C}(0,\rho)} \frac{e^{b_0 z} \left(e^{b_2 z} - 1\right)^{n-k}}{z^{n+1}} \, dz. \tag{14}$$

One can do is, use similar thing what Corcino [10] estimate the integral, for $r - Whitney$ numbers.

**Theorem 2.4.** *Let $(T(n, k))$ be sequence satisfying the usual initial condition (1) and*

$$T(n, k) = a_0 \, T(n-1, k) + (b_2 n - b2k + b_0) \, T(n-1, k-1).$$

*Let $\rho^* > 0$ be the unique solution of the saddle-point equation*

$$\rho^* \left( b_0 + \frac{(n-k) b_2}{1 - e^{-b_2 \rho^*}} \right) = n. \tag{15}$$

*Define*

$$B = \frac{1}{2} \left[ \frac{b_0}{n-k} + \frac{b_2 e^{b_2 \rho^*} (e^{b_2 \rho^*} - b_2 \rho^* - 1)}{(e^{b_2 \rho^*} - 1)^2} \right]. \tag{16}$$

*Then, for $\frac{b_0}{b2} \geq 4/3$,*

$$T(n, k) \sim \frac{n!}{(n-k)!} \frac{a_0^{n-k}}{b_2^{n-k}} \frac{e^{b_0 \rho^*} (e^{b_2 \rho^*} - 1)^{n-k}}{2 \, (\rho^*)^n \sqrt{\pi (n-k) \rho^* B}} (1 + s), \tag{17}$$

*with*

$$s = \frac{1}{(n-k)\,\rho^*}\Bigg\{\frac{3}{4\rho^* B^2}\Bigg[\frac{b_0\rho^*}{24(n-k)} + \frac{1}{24}\Big(b_2\rho^* + (b_2\rho^* - 7(b_2\rho^*)^2 + 6(b_2\rho^*)^3 - (b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-1}\Big)$$
$$+\frac{1}{24}\Big((18(b_2\rho^*)^3 - 7(b_2\rho^*)^2 - 7(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-2}\Big)$$
$$+ (12(b_2\rho^*)^3 - 12(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-3} - \frac{1}{24}\Big(6(b_2\rho^*)^4(e^{b_2\rho^*}-1)^{-4}\Big)\Bigg]$$
$$-\frac{15}{16(\rho^*)^2B^3}\Bigg[\frac{b_0\rho^*}{6(n-k)} + \frac{1}{6}\Big(b_2\rho^* + (b_2\rho^* - 3(b_2\rho^*)^2 + (b_2\rho^*)^3)(e^{b_2\rho^*}-1)^{-1}\Big)$$
$$+ (3(b_2\rho^*)^3 - 3(b_2\rho^*)^2)(e^{b_2\rho^*}-1)^{-2} + \frac{1}{6}\Big(2(b_2\rho^*)^3(e^{b_2\rho^*}-1)^{-3}\Big)\Bigg]^2\Bigg\}.$$

*Also, it is still true even* $q := \frac{b_0}{b2} < 4/3$*, if* $k \le n/q$*.*

We note that the $\rho^*$ at (15) is nothing else than $\zeta_{n-k}(n)$, according to (12).

*Proof.* From (14),

$$T(n,k) = \frac{n!}{2\pi i}\frac{a_0^{n-k}}{(n-k)!b_2^{n-k}}I, \quad I = \int_{\mathcal{C}(0,R)} \frac{e^{b_0 z}\big(e^{b_2 z}-1\big)^{n-k}}{z^{n+1}}\,dz. \tag{18}$$

We parametrize the contour by

$$z = \rho e^{i\theta}, \qquad dz = i\rho e^{i\theta} d\theta.$$

Substituting into the integral gives

$$I = \int_{-\pi}^{\pi} \frac{e^{b_0\rho e^{i\theta}}\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k}}{(\rho e^{i\theta})^{n+1}}\, i\rho e^{i\theta}\,d\theta = i\rho^{-n}\int_{-\pi}^{\pi} \exp\big(b_0\rho e^{i\theta}\big)\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k} e^{-in\theta}\,d\theta.$$

Thus,

$$I = i\rho^{-n}\int_{-\pi}^{\pi}\exp(h_\rho(\theta))\,d\theta, \quad h_\rho(\theta) = b_0\rho e^{i\theta} + (n-k)\,\log\big(e^{b_2\rho e^{i\theta}}-1\big) - in\theta.$$

The next step is expand $h_\rho(\theta)$ and choose $\rho^*$, saddle point ( at $\theta = 0$): we choose is as radius of the countour. So in the expansion, the order zero is constant (can put out of the integral); the order is zero by definition.We have then

$$T(n,k) = (\textbf{constant}) \times \int_{-\pi}^{\pi}\exp\big(-(n-k)B\theta^2\big)\exp\big(\textbf{oder superior to 3}\big).$$

Now, we want to bring it to gamma function, so we need $x^2 = (n-k)B\theta^2$.
The next task is to argue one the domain of integral, so it can extend to infinity and many computation to have the result. For many details of these step, please see [11],. □

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp\bigg(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\bigg). \tag{19}$$

We can also see that $F_x(t) = F(t,x)$ is H-admissible for $x > 0$ with

$$\rho = +\infty, \quad a(r) = r\left(b_0x + a_0e^{b_2xr}\right), \quad b(r) \;= a_0b_2x\,r^2e^{b_2xr} + a_0re^{b_2xr} + b_0x\,r.$$

Then Coefficients of admissible functions theorem ensures us

$$[t^n]F(t,x) \sim \frac{\exp\Big(b_0 x\zeta_x(n) + \frac{a_0}{b_2 x}\left(e^{b_2 x\zeta_x(n)} - 1\right)\Big)}{\zeta_x(n)^n\sqrt{2\pi\left(a_0 b_2 x\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)} + n\right)}} \tag{20}$$

where $\zeta_x(n)\left(b_0 x + a_0 e^{b_2 x\zeta_x(n)}\right) = n$.

We want to try to use the Quasi-Powers, general distributions: Laplace transform

$$\lambda_n(s) = \mathbb{E}(e^{sX_n}) = \frac{[t^n]F(t,e^s)}{[t^n]F(t,1)},$$

is of form

$$\lambda_n(s) = \exp\big(\beta_n U(s) + V(s)\big)\left(1 + O\Big(\frac{1}{\kappa_n}\Big)\right),$$

with $\beta_n, \kappa_n \to +\infty$ as $n \to +\infty$, and $U''(0) \neq 0$. $U(s)$ and $V(s)$ is analytic near 0.

I tried but I don't see so far... after many try...

Now, let's try the Generalized quasi-Powers theorem. We need to write $p_n(x) = \exp(h_n(t,x))$. We need to start by approximating $p_n(x)$ near $x = 1$. If $h(t,x) = b_0 xt + \frac{a_0}{b_2 x}\left(e^{b_2 xt} - 1\right)$ then $F(t,x) = \exp(h(t,x))$ and the saddle point equation is

$$\frac{\partial}{\partial t}h(t,x)\Big|_{t=\zeta_x(n)} = \frac{n}{\zeta_x(n)} \implies \zeta_x(n)\Big(b_0 x + a_0 e^{b_2 x\zeta_x(n)}\Big) = n. \tag{21}$$

We get now (20) and compute

$$p_n(x) \sim \frac{F(\zeta_x(n),x)}{F(\zeta_1(n),1)}\Big(\frac{\zeta_1(n)}{\zeta_x(n)}\Big)^n K_n(x)(1+o(1)), \quad K_n(x) = \sqrt{\frac{n + a_0 b_2\zeta_1(n)^2 e^{b_2\zeta_1(n)}}{n + a_0 b_2 x\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}}.$$

The $h_n(x)$ wanted is

$$h_n(x) = \log p_n(x) = h(\zeta_x(n),x) - h(\zeta_1(n),1) + n\log\Big(\frac{\zeta_1(n)}{\zeta_x(n)}\Big) + \log K(x) + o(1).$$

Checking carefully, on has

$$\frac{h_n'''(x)}{\big(h_n'(1) + h_n''(1)\big)^{3/2}} \to 0 \quad (n \to \infty).$$

Now, from the Generalized quasi-Powers theorem, we can conclude that the associated $X_n$ converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1) + h_n''(1)$.
Using elementary calculus, chain rule with equation (21),

$$h_n'(x) = \partial_x h(\zeta_x(n),x) + K_n'(x)/K_n(x). \tag{22}$$

Then

$$\mu_n \sim h_n'(1) \sim b_0\zeta_1(n) + a_0\zeta_1(n)e^{b_2\zeta_1(n)} - \frac{a_0}{b_2}(e^{b_2\zeta_1(n)} - 1) + K_n'(1)/K_n(1). \tag{23}$$

If $D(x) = n + a_0 b_2 x\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}$, $K_n'(x)/K_n(x) = -\frac{1}{2}\frac{D'(x)}{D(x)}$. Then

$$\frac{K'(x)}{K(x)} = -\frac{1}{2}\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}\Big(\zeta_x(n)^2 + 2x\,\zeta_x(n)\,\partial_x\zeta_x(n) + b_2 x\,\zeta_x(n)^3 + b_2 x^2\,\zeta_x(n)^2\,\partial_x\zeta_x(n)\Big)}{n + a_0 b_2 x\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}},$$

with

$$\partial_x\zeta_x(n) = -\frac{b_0\,\zeta_x(n) + a_0 b_2\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2\,\zeta_x(n)e^{b_2 x\zeta_x(n)}}.$$

So,

$$\frac{K'(x)}{K(x)} = -\frac{1}{2}\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}}{n + a_0 b_2 x\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}\Bigg[\zeta_x(n)^2 + b_2 x\,\zeta_x(n)^3 - \left(2x\,\zeta_x(n) + b_2 x^2\,\zeta_x(n)^2\right)\frac{b_0\,\zeta_x(n) + a_0 b_2\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2 x\,\zeta_x(n) e^{b_2 x\zeta_x(n)}}\Bigg].$$

As asymptotically, we have $b_0 x + a_0 e^{b_2\zeta_x(n)} \sim a_0 e^{b_2\zeta_x(n)}$, the saddle point equation gives us

$$\zeta_x(n) a_0 e^{b_2\zeta_x(n)} \sim n.$$

First, the prefactor is $\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}}{n + a_0 b_2 x\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}} \sim \frac{1}{\zeta_x(n) + x\zeta_x(n)^2} \sim \frac{1}{x\zeta_x(n)^2}$. For the fraction in bracket, exponential terms dominate: $\frac{b_0\zeta_x(n) + a_0 b_2\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2 x\zeta_x(n) e^{b_2 x\zeta_x(n)}} \sim \frac{\zeta_x(n)}{x}$. So the bracket: $\zeta_x(n)^2 + b_2 x\zeta_x(n)^3 - (2x\zeta_x(n) + b_2 x^2\zeta_x(n)^2)\frac{\zeta_x(n)}{x} = -\zeta_x(n)^2$. Finally, $\frac{K'(x)}{K(x)} \sim -\frac{1}{2}\cdot\frac{1}{x\zeta_x(n)^2}\cdot(-\zeta_x(n)^2) = \frac{1}{2x}$. Therefore (23) implies

$$\mu_n \sim n - \frac{a_0}{b_2} e^{b_2\zeta_1(n)}. \tag{24}$$

To estimate $\frac{a_0}{b_2} e^{b_2\zeta_1(n)}$, let $y_n = \zeta_1(n)$. Then asymptotically (21) becomes

$$y_n a_0 e^{b_2 y_n} \sim n. \tag{25}$$

Taking logarithm both sides, we have $\log(y_n) + \log(a_0) + b_2 y_n \sim \log(n)$. Noting that $y_n$ go to infinity as $n \to \infty$, $b_2 y_n \sim \log(y_n) + \log(a_0) + b_2 y_n \sim \log(n)$. Then

$$y_n \sim \frac{\log(n)}{b_2}. \tag{26}$$

So (25) implies $\frac{a_0}{b_2} e^{b_2 y_n} \sim \frac{1}{b_2} \times \frac{n}{y_n} \sim \frac{n}{\log(n)}$. Thus (24) gives us

$$\mu_n \sim n - \frac{n}{\log(n)}.$$

If we want more precision than (26), the equation (25) can give us $b_2 y_n e^{b_2 y_n} \sim \frac{b_2 n}{a_0}$, that means $b_2 y_n = W(\frac{b_2 n}{a_0})$. So $y_n \sim \frac{W(\frac{b_2 n}{a_0})}{b_2}$. With (25) again, $\frac{a_0}{b_2} e^{b_2 y_n} \sim \frac{1}{b_2} \times \frac{n}{y_n} \sim \frac{n}{W(\frac{b_2 n}{a_0})}$. Thus (24) gives us

$$\mu_n \sim n - \frac{n}{W(\frac{b_2 n}{a_0})} + \frac{a_0}{b_2}.$$

For the same reasoning as we did above for $\frac{K'(x)}{K(x)}$, we got also similar thing for its derivative. So from (22)

$$h_n''(x) \sim \partial_x\zeta_x(n) h_t(\zeta_x(n), x) + h_x x(\zeta_x(n), x).$$

After computation, we get

$$h_n''(1) = \frac{2a_0}{b_2}\left(e^{b_2 y_n} - 1\right) - 2a_0 y_n e^{b_2 y_n} + a_0 b_2 y_n^2 e^{b_2 y_n} - \frac{y_n^2\left(b_0 + a_0 b_2 y_n e^{b_2 y_n}\right)^2}{n + a_0 b_2 y_n^2 e^{b_2 y_n}}.$$

Then

$$\begin{aligned} h_n'(1) + h_n''(1) &\sim n + \frac{a_0}{b_2}\left(e^{b_2 y_n} - 1\right) - 2n + nW(\frac{b_2 n}{a_0}) - \frac{\left(nW(\frac{b_2 n}{a_0})\right)^2}{n\left(1 + W(\frac{b_2 n}{a_0})\right)} \\ &\sim -n + \frac{n}{W(\frac{b_2 n}{a_0})} + nW(\frac{b_2 n}{a_0}) - \frac{nW(\frac{b_2 n}{a_0})^2}{\left(1 + W(\frac{b_2 n}{a_0})\right)} - \frac{a_0}{b_2} \\ &\sim \frac{n}{W(\frac{b_2 n}{a_0})\left(1 + W(\frac{b_2 n}{a_0})\right)} - \frac{a_0}{b_2}. \end{aligned}$$

After computation

$$h_n'''(x) = -\frac{6a_0}{b_2x^4}\left(e^{b_2x\zeta_x(n)} - 1\right)$$
$$+\frac{6a_0\zeta_x(n)}{x^3}e^{b_2x\zeta_x(n)} - \frac{3a_0b_2\zeta_x(n)^2}{x^2}e^{b_2x\zeta_x(n)} + \frac{a_0b_2^2\zeta_x(n)^3}{x}e^{b_2x\zeta_x(n)} + 2a_0b_2^2\zeta_x(n)^2e^{b_2x\zeta_x(n)}\,\partial_x\zeta_x(n)$$
$$+ a_0b_2e^{b_2x\zeta_x(n)}\Big(1 + b_2x\zeta_x(n)\Big)\big(\partial_x\zeta_x(n)\big)^2 + \Big(b_0 + a_0b_2\zeta_x(n)e^{b_2x\zeta_x(n)}\Big)\partial_{xx}\zeta_x(n) + \left(\frac{K'}{K}\right)''(x).$$

Noticing $\zeta_x(n) \sim \dfrac{\log n}{b_2x}$, $\partial_x\zeta_x(n) \sim -\dfrac{\log n}{b_2x^2}$ and $\partial_{xx}\zeta_x(n) \sim \dfrac{2\log n}{b_2x^3}$, we get asymptotically $h_n'''(x) \sim \frac{2n\log(n)}{x^3}$ and

$$\frac{h_n'''(x)}{\big(h_n'(1) + h_n''(1)\big)^{3/2}} \sim \frac{2}{x^3}\frac{(\log(n))^4}{\sqrt{n}} \to 0,$$

uniformly for $|x-1| < \frac{1}{2}$.
Therefore we have the following proposition.

**Proposition 2.3.** *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = a_0\,T(n-1,k) + (b_2n - b2k + b_0)\,T(n-1,k-1).$$

*Assume that all coefficients are non-negative (also all $a_i, b_i$). $F(t,x)$ be the exponential bivariate generating function. Then The random variable $X_n$ of probability generating function*

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

*converges in distribution to a a Gaussian with mean and variance*

$$\mu_n \sim n - \frac{n}{W(\frac{b_2n}{a_0})} + \frac{a_0}{b_2}; \qquad \sigma_n^2 \sim \frac{n}{W(\frac{b_2n}{a_0})\big(1 + W(\frac{b_2n}{a_0})\big)} - \frac{a_0}{b_2}.$$

| $n$ | $E(X_n)$ | $\mu_n$ | Relative error | $V(X_n)$ | $\sigma_n^2$ | Relative error |
|---|---|---|---|---|---|---|
| 50 | 30.327051 | 29.526468 | 0.02640 | 4.757491 | 4.654462 | 0.02166 |
| 100 | 65.795499 | 64.928556 | 0.01318 | 7.757654 | 7.672991 | 0.01091 |
| 200 | 140.401692 | 139.480597 | 0.00656 | 12.572785 | 12.503131 | 0.00554 |
| 500 | 374.724496 | 373.746366 | 0.00261 | 23.814637 | 23.760281 | 0.00228 |
| 1000 | 778.299820 | 777.286986 | 0.00130 | 38.832127 | 38.786608 | 0.00117 |
| 1200 | 942.032926 | 941.011941 | 0.00108 | 44.216357 | 44.172846 | 0.00098 |
| 1500 | 1189.234172 | 1188.203701 | 0.00087 | 51.873840 | 51.832631 | 0.00079 |
| 1900 | 1521.115543 | 1520.075571 | 0.00068 | 61.499379 | 61.460433 | 0.00063 |
| 2000 | 1604.416500 | 1603.374536 | 0.00065 | 63.822328 | 63.783852 | 0.00060 |
| 2500 | 2022.525466 | 2021.475114 | 0.00052 | 75.038569 | 75.002046 | 0.00049 |

Table 2: Comparison of exact and approximate values with relative errors for $a_0 = 8, b_0 = 4, b_2 = 3$.

## 2.5 Case2A: $b_1 + b_2 \neq 0$, $a_2 = 0$ and $b_1 \neq 0$ ($b_1 < 0$)

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = (1 - (b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times\exp\left[-\frac{a_0}{b_1x}\left((1 - (b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}} - 1\right)\right].$$

Note that the singularity at $x=0$ is removable.
It is convenient to write $F(t,x)=\exp(h(t,x))$, where

$$h(t,x)=-\frac{b_0+b_1+b_2}{b_1+b_2}\log\bigl(1-(b_1+b_2)xt\bigr)-\frac{a_0}{b_1x}\Bigl((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}-1\Bigr).$$

The saddle point $\zeta_x(n)$ is defined implicitly by

$$\frac{\partial}{\partial t}h(t,x)\Big|_{t=\zeta_x(n)}=\frac{n}{\zeta_x(n)}.$$

So

$$\frac{(b_0+b_1+b_2)x}{1-(b_1+b_2)x\zeta_x(n)}+a_0\left(1-(b_1+b_2)x\zeta_x(n)\right)^{-\frac{b_2}{b_1+b_2}}=\frac{n}{\zeta_x(n)}.$$

We know that the saddle point approach to dominant singularity as $n\to\infty$, let's rewrite the above equation as

$$\frac{1}{(1-(b_1+b_2)x\zeta_x(n))^{\frac{b_2}{b_1+b_2}}}\left((1-(b_1+b_2)x\zeta_x(n))^{-\frac{b_1}{b_1+b_2}}+a_0\right)=\frac{n}{\zeta_x(n)}.$$

So asymptotically,

$$1-(b_1+b_2)x\,\zeta_x(n)\sim\left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}}.$$

So

$$\zeta_x(n)=\frac{1}{(b_1+b_2)x}\left(1-\left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}}+o\Bigl(n^{-\frac{b_1+b_2}{b_2}}\Bigr)\right).$$

We can check that $F(z,x)$ is H-admissible, the coefficient extraction admits the classical saddle–point estimate

$$[t^n]F(t,x)\sim\frac{F(\zeta_x(n),x)}{\zeta_x(n)^n\sqrt{2\pi h''(\zeta_x(n),x)}}.$$

So one obtains

$$p_n(x)\sim\frac{F(\zeta_x(n),x)}{F(\zeta_1(n),1)}\Bigl(\frac{\zeta_1(n)}{\zeta_x(n)}\Bigr)^nK_n(x)(1+o(1)),\qquad K_n(x)=\sqrt{\frac{n+\zeta_1(n)^2h''(\zeta_1(n),1)}{n+\zeta_x(n)^2h''(\zeta_x(n),x)}}.$$

Now, I want to apply the Generalized quasi-powers, we need $p_n=\exp(h_n)$, so

$$h_n(x)=\log p_n(x)=h(\zeta_x(n),x)-h(\zeta_1(n),1)+n\log\frac{\zeta_1(n)}{\zeta_x(n)}+\log K_n(x)+o(1).$$

Checking carefully, on has

$$\frac{h_n'''(x)}{\bigl(h_n'(1)+h_n''(1)\bigr)^{3/2}}\to 0\quad(n\to\infty).$$

So the Generalized quasi-powers theorem implies that the random variable associated $X_n$ converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1)+h_n''(1)$.

..... COMPUTATION TO BE CHECK CAREFULLY AND CONTINUED... SOME CONDITIONS REQUIRED!!!

## 2.6 Case 3A: $b_1, a_2, b_1 + b_2 \neq 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times \left(1 - \frac{a_2}{b_1 x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]\right)^{-\frac{a_0+a_2}{a_2}}.$$

Note that the singularity at $x = 0$ is removable. We want to use Algebraic singularity schema, meanly we try to write

$$F(t,x) = A(t,x) + B(t,x)C(t,x)^{-\alpha}. \tag{27}$$

$A = 0, B(t,x) = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$ and $C(t,x) = 1 - \frac{a_2}{b_1 x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]$. Algebraic perturbation is verified for good choice of $r < \frac{1}{b_1+b_2}$. Under some conditions on the sign $a_i, b_i$, non-degeneracy is also verified... to write in details...

$$\rho(x) = \frac{1}{(b_1+b_2)x}\Big(1-\big(1-\frac{b_1}{a_2}x\big)^{\frac{b_1+b_2}{b_1}}\Big),$$

whose the singularity at $x = 0$ is removable.

..... AFTER computation (TO TAP LATER),

We see that

$$\mathfrak{v}\left(\frac{\rho(1)}{\rho(u)}\right) = -\frac{\rho''(1)}{\rho(1)} - \frac{\rho'(1)}{\rho(1)} + \left(\frac{\rho'(1)}{\rho(1)}\right)^2 \neq 0.$$

So the random variable $X_n$ of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a Gaussian with speed of convergence that is $O(n^{-\frac{1}{2}})$. The mean $\mu_n$ and the standard deviation $\sigma_n^2$ are asymptotically linear in $n$. More precisely $\mu_n = \mathfrak{m}\big(\frac{\rho(1)}{\rho(x)}\big)n$ and and $\sigma_n^2 = \mathfrak{v}\left(\frac{\rho(1)}{\rho(u)}\right)n$. Noting $\mathfrak{m}\big(\frac{\rho(1)}{\rho(x)}\big) = -\frac{\rho'(1)}{\rho(1)}$, we have explicitly

$$\mu_n = \left(1 - \frac{b_1+b_2}{a_2}\cdot\frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-1}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}}\right)n,\ \sigma_n^2 = \left[\frac{(b_1+b_2)b_2}{a_2^2}\cdot\frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-2}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}} + \left(\frac{b_1+b_2}{a_2}\cdot\frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-1}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}} - 1\right)^2\right]n.$$

... CONDITIONs ON PARAMETERS WILL BE VERIFIED// CHECKED WITH THE COMPUTATIONS...

# References


[1] D. Dumont, "William Chen grammars and derivations in trees and arborescences," *Séminaire Lotharingien de Combinatoire*, vol. 37, pp. B37a–21, 1996.

[2] P. Flajolet and R. Sedgewick, Analytic Combinatorics, *Cambridge University Press*, 2009. Available: https://doi.org/10.1017/CBO9780511801655

[3] H.-K. Hwang, H.-H. Chern, and G.-H. Duh, An asymptotic distribution theory for Eulerian recurrences with applications, *Advances in Applied Mathematics*, 112, Article 101960, 2020.

[4] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed., Addison-Wesley, Reading, MA, 1994.

[5] E. Neuwirth, Recursively defined combinatorial functions: extending Galton's board, *Discrete Mathematics*, 239(1), 33–51, 2001. doi:10.1016/S0012-365X(00)00373-3.

[6] M. Z. Spivey, On Solutions to a General Combinatorial Recurrence, *Journal of Integer Sequences*, 14, Article 11.9.7, 2011. Available: https://cs.uwaterloo.ca/journals/JIS/VOL14/Spivey/spivey31.pdf

[7] H. S. Wilf, The method of characteristics, and "problem 89" of Graham, Knuth and Patashnik, arXiv:math/0406620, 2004. https://arxiv.org/abs/math/0406620

[8] P. Hitczenko, On limiting distributions of Graham–Knuth–Patashnik recurrences, *Advances in Applied Mathematics*, 2020. (Note: verify volume and page numbers.)

[9] B. Randrianirina, *Espèces mixtes et système d'équations différentielles*. Annales des Sciences Mathématiques du Québec, vol. 24, no. 2, pp. 179–211, 2000, available online at `https://www.labmath.uqam.ca/annales/indexgeneral.html`.

[10] C. B. Corcino, R. B. Corcino, and N. Acala, Asymptotic Estimates for r-Whitney Numbers of the Second Kind, *Journal of Applied Mathematics*, vol. 2014, Article ID 354053, 2014. Available: https://doi.org/10.1155/2014/354053

[11] C. B. Corcino and R. B. Corcino, Asymptotic Estimates for Second Kind Generalized Stirling Numbers, *Journal of Applied Mathematics*, vol. 2013, Article ID 918513, 2013. Available: https://doi.org/10.1155/2013/918513

[12] C. B. Corcino and R. B. Corcino, "Asymptotic estimates for second kind generalized Stirling numbers," *Journal of Applied Mathematics*, vol. 2013, Article ID 918513, 7 pages, 2013.